Monte Carlo simulation of magnetic nanodots on graphics hardware: a single fused kernel for the extended Heisenberg chain

This section carries the simulator that produces the spin configurations used throughout the remainder of the work. It states the lattice model and the multi-sublattice Metropolis chain it borrows, then defines the proposal: the realisation of one pass of that chain as a single fused kernel, built from four components named for the properties they deliver — contiguous neighbour access, a synchronisation-free halo, counter-based proposals and deterministic observables. The simulator is validated against a reference implementation of the same chain under identical randomness, and its rate is reported per single-site update against the processor baseline, which runs the same code and the same Hamiltonian on the same workload. At the production workload — 200 temperatures by 20 000 sweeps for one hundred states at five thousand replicas — it completes in 57.6 min on four datacentre devices against 130.7 d on the processor baseline, a separation of 834-fold per site update on identical physics.

1 The model and the chain

This section states the two pieces of existing theory the simulator borrows: the lattice model and the multi-sublattice Metropolis chain. The proposal — the realisation of one pass of that chain as a single fused kernel — follows in Section 2.

1.1 The lattice model

Let $\Lambda \subset \mathbb{Z}^3$ be the set of lattice sites of a cylindrical nanodot,

$$\Lambda = \{(x, y, z) \in \mathbb{Z}^3 : x^2 + y^2 \le R_d^2,\ 0 \le z < L\}, \tag{1}$$

where R_d is the dot radius in lattice steps and L the number of layers. Writing $L_2 = \lfloor R_d \rfloor + 1$ and $N = 2L_2 + 1$, the sites are addressed inside a box of $L \times N \times N$, of which $|\Lambda|$ lie inside the disc. Because N is odd by construction, the box extent carries no factor of two — a fact the proposal uses in Section 2.

Two words are used throughout in a fixed sense and are declared here, because the literature uses one of them differently. A *state* is one point θ in the parameter space of the Hamiltonian below — the temperature, the three outer exchange constants, the two anisotropy constants, the field and the chiral constant — and is the unit a production run is counted in: a hundred states means a hundred such parameter points, each simulated down its own temperature ladder. A *replica* is one independent Markov chain at the same θ, so replicas of one state differ only in their randomness and are averaged over. Where the accelerated-Monte-Carlo literature says "state" it usually means the value a spin takes, which here is never called that.

To each site $i \in \Lambda$ is attached a classical spin $\mathbf{s}_i \in \mathbb{R}^3$ with $\|\mathbf{s}_i\| = S_0$; the configuration is $\mathbf{s} = \{\mathbf{s}_i\}_{i \in \Lambda}$. Let $\mathcal{N}(i) = \{i + \mathbf{d} : \mathbf{d} \in \mathcal{D}_k,\ k \in \mathcal{S}\} \cap \Lambda$ be the interacting neighbourhood of i, and let $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ be the lattice directions. The in-plane directions wrap; z does not, so the top and bottom layers have no neighbour beyond them.

The energy has five terms:

$$\begin{aligned}\mathcal{H}(\mathbf{s}) = &-\sum_{k\in\mathcal{S}} J_k \sum_{\langle ij\rangle_k} \mathbf{s}_i\cdot\mathbf{s}_j \;+\; \sum_i \mathcal{A}_i(\mathbf{s}_i) \\ &-\; D\sum_i \Big(\big[\mathbf{s}_i\times\mathbf{s}_{i+\hat{\mathbf{x}}}\big]_y - \big[\mathbf{s}_i\times\mathbf{s}_{i+\hat{\mathbf{y}}}\big]_x \Big) \;-\; \mathbf{H}\cdot\sum_i \mathbf{s}_i .\end{aligned} \tag{2}$$

The first term is the exchange over the active shells $\mathcal{S}\subseteq\{1,2,3,4\}$, where $\langle ij\rangle_k$ runs once over each bond of shell k and $\mathcal{D}_k$ is that shell's displacement set: the six displacements at squared distance 1, the twelve at 2, the eight at 3 and the six at 4. The first-shell constant is normalised, $J_1 = 1$, and sets the energy scale. The Dzyaloshinskii–Moriya term is *interfacial*: it runs over in-plane bonds only, with the sign shown. That restriction is a modelling commitment, not a simplification — the all-axes form is a different physical system, and on a common configuration the two give total energies that differ by tens of percent.

The anisotropy $\mathcal{A}_i$ is the cubic series written on direction cosines, with a site-dependent first constant:

$$\mathcal{A}_i(\mathbf{s}) = \frac{K_i^{\text{eff}}}{S_0^4}\big[(s_x s_y)^2 + (s_x s_z)^2 + (s_y s_z)^2\big] + \frac{K_2}{S_0^6}(s_x s_y s_z)^2, \qquad K_i^{\text{eff}} = \begin{cases} K_S & i\in\partial\Lambda, \\ K_1 & \text{otherwise,}\end{cases} \tag{3}$$

where $\partial\Lambda$ is the set of sites with fewer than six in-disc first neighbours. Surface anisotropy therefore enters as a *field over sites*, not as a scalar, and that is why it is carried as an array of the same shape as one spin component throughout.

1.2 The multi-sublattice Metropolis chain

Let $\beta = 1/k_B T$. A single-site Metropolis step at site i draws a trial spin $\mathbf{t}_i$ uniformly on the sphere of radius S_0, forms $\Delta\mathbf{s}_i = \mathbf{t}_i - \mathbf{s}_i$, and accepts with probability $\min\{1, e^{-\beta\Delta E_i}\}$, where ΔE_i is the energy difference the single-site change produces. Because every term of (2) couples i only to $\mathcal{N}(i)$ and to itself, ΔE_i reads

$$\Delta E_i = -\,\Delta\mathbf{s}_i\cdot\sum_{k\in\mathcal{S}} J_k \sum_{\mathbf{d}\in\mathcal{D}_k} \mathbf{s}_{i+\mathbf{d}} + \mathcal{A}_i(\mathbf{t}_i) - \mathcal{A}_i(\mathbf{s}_i) - D\,\delta_i^{\text{DM}} - \mathbf{H}\cdot\Delta\mathbf{s}_i, \tag{4}$$

with δ_i^{DM} the in-plane cross-product difference of the DMI term. The DMI term stays on first-shell bonds whatever else is active, because it is interfacial by construction. The quantity that matters for what follows is the *weighted neighbour sum*

$$\mathbf{v}_i = \sum_{k\in\mathcal{S}} J_k \sum_{\mathbf{d}\in\mathcal{D}_k} \mathbf{s}_{i+\mathbf{d}}; \tag{5}$$

it is the only part of ΔE_i that reaches outside site i.

Let $\gamma : \Lambda \to \{0,\dots,n_c-1\}$ be a colouring of the lattice, with colour classes $\Lambda^{(c)} = \gamma^{-1}(c)$ so that $\Lambda = \bigsqcup_c \Lambda^{(c)}$. The colouring is *admissible* for the active shells when no displacement of any active shell preserves it,

$$\gamma(i) \neq \gamma(i+\mathbf{d}) \qquad \text{for every } \mathbf{d}\in\mathcal{D}_k,\ k\in\mathcal{S}. \tag{6}$$

Under an admissible colouring, ΔE_i for $i\in\Lambda^{(c)}$ depends on no spin of $\Lambda^{(c)}$, so the sites of one colour may be updated *simultaneously* without changing the chain: the simultaneous update is an exact reordering of the sequential sweep, not an approximation of it. One sweep is n_c passes, $c = 0$ through $n_c - 1$.

Admissibility is decided, shell by shell, by what each displacement does to the parity of $x + y + z$. The shells are the displacement sets at squared distance 1, 2, 3 and 4, and each has a different arithmetic:

- **Shell 1**, $\mathcal{D}_1 = \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\}$, six displacements. Exactly one coordinate changes by one, so $x+y+z$ changes by ± 1 and the parity *flips*. Every first neighbour of a site is therefore of the other colour.
- **Shell 2**, the twelve permutations of $(\pm 1, \pm 1, 0)$. Two coordinates change by one, so the sum changes by 0 or ± 2 and the parity is *preserved*: a second neighbour sits in the same class as the site itself, and a two-colouring is no longer admissible.
- **Shell 3**, the eight sign choices of $(\pm 1, \pm 1, \pm 1)$. All three coordinates change by one, so the sum changes by an odd number and the parity flips again. Shell 3 is compatible with the two-colouring, and $\{1, 3\}$ together still are.
- **Shell 4**, the six displacements $(\pm 2, 0, 0)$ and permutations. One coordinate changes by *two*. The parity is preserved, and worse, the displacement is invisible to any colouring built from residues modulo two, because x and $x \pm 2$ share every such residue. Stride three on all three axes is the narrowest scheme that separates it.

So the colouring widens for an arithmetic reason, not an arbitrary one, and it widens in two steps: a second neighbour forces residues modulo two on each axis separately, giving $2^3 = 8$ classes, and a fourth neighbour forces residues modulo three, giving $3^3 = 27$. Table 1 collects them.

Table 1: The narrowest admissible colouring for each combination of active shells. The cost of a wider colouring falls on the parallelism and not on the work: a sweep performs $|\Lambda|$ site updates whatever n_c is, but each pass performs only $|\Lambda|/n_c$ of them concurrently.

active shells $\mathcal{S}$	$\gamma(x, y, z)$	n_c
$\{1\}$ or $\{1, 3\}$	$(x + y + z) \bmod 2$	2
any set containing 2, none containing 4	$(x \bmod 2,\ y \bmod 2,\ z \bmod 2)$	8
any set containing 4	$(x \bmod 3,\ y \bmod 3,\ z \bmod 3)$	27

The colouring in use is *verified* against (6) over the displacement sets of the active shells rather than taken from Table 1, so a combination the table does not anticipate is detected instead of assumed.

2 The fused kernel

The chain of Section 1.2 is the object being realised, and the proposal is the realisation of one of its passes as a single fused kernel. It is defined by four components, each named for the property it delivers, and it computes the weighted neighbour sum $\mathbf{v}_i$ of (5) without ever materialising a translate of the configuration.

2.1 The matrix decomposition

Before any layout question, the chain of Section 1.2 is written as operations on arrays, because that is the form the kernel realises. Let B be the number of replicas advanced together and let

$$S \in \mathbb{R}^{B \times 3 \times L \times N \times N}, \qquad S[b, a, z, y, x] = (\mathbf{s}_i)_a \ \text{ for } i = (x, y, z) \text{ of replica } b, \tag{7}$$

so the component index $a \in \{0, 1, 2\}$ is an axis of the array and not a field of a record. Sites outside the disc occupy their place in the box and are excluded by a mask, which is cheaper than an index list and keeps every array rectangular.

The translate. For a displacement $\mathbf{d} = (d_x, d_y, d_z)$ define the translate $T_{\mathbf{d}}$ by

$$(T_{\mathbf{d}}S)[b, a, z, y, x] = S[b, a, z + d_z,\ (y + d_y) \bmod N,\ (x + d_x) \bmod N], \tag{8}$$

with the in-plane indices wrapping and the out-of-plane one *gated*: a layer displaced past $z = 0$ or $z = L - 1$ contributes zero rather than wrapping, which is the open boundary of Section 1.1 expressed on the array. Each $T_{\mathbf{d}}S$ has the same shape as S.

The neighbour sum. The weighted neighbour sum of (5) becomes one array of the shape of S,

$$V = \sum_{k \in \mathcal{S}} J_k \sum_{\mathbf{d} \in \mathcal{D}_k} T_{\mathbf{d}}S, \tag{9}$$

a sum of $\sum_k |\mathcal{D}_k|$ translates — six for the first shell alone, twenty-six when all four are active.

The proposal and the energy difference. Let T hold the trial spins, of the shape of S, and $\Delta = T - S$. Contracting the component axis turns the single-site energy difference (4) into one array with the component axis removed,

$$\Delta E \in \mathbb{R}^{B \times L \times N \times N}, \qquad \Delta E = -\sum_a \Delta[\,\cdot\,, a] \odot V[\,\cdot\,, a] + \mathcal{A}(T) - \mathcal{A}(S) - D\,\delta^{\mathrm{DM}} - \sum_a H_a\,\Delta[\,\cdot\,, a], \tag{10}$$

where $\odot$ is the elementwise product. The anisotropy $\mathcal{A}$ is elementwise in the three components and carries K^{eff}, itself an array of shape $B \times L \times N \times N$ rather than a scalar, which is how the surface constant of (3) enters. The chiral term δ^{DM} needs only the $\pm\hat{\mathbf{x}}$ and $\pm\hat{\mathbf{y}}$ translates already formed for V.

The update. With $M^{(c)}$ the boolean mask of colour class c and U an array of uniforms, both of shape $B \times L \times N \times N$, one pass is a single masked select,

$$S \ \leftarrow\ \mathrm{where}\Big(\underbrace{M^{(c)} \wedge [\Delta E \le 0 \ \vee\ U < e^{-\beta \Delta E}]}_{\text{accept, broadcast over } a},\ T,\ S\Big). \tag{11}$$

The acceptance test is decided once per site and then broadcast along the component axis, so a spin is replaced in all three components or in none — which is what makes the trial a rotation of the whole vector rather than three independent moves.

Equations (7)–(11) are the specification. Read literally they would allocate one array of the shape of S per translate, and that allocation is exactly what the kernel of Sections 2.2–2.4 avoids: it forms V in registers and writes only the result of (11). The remaining subsections say how.

2.2 Contiguous neighbour access: the component-planar layout

Let B be the number of independent replicas advanced together. The configuration tensor may be laid out component-minor, with shape $(B, L, N, N, 3)$, or *component-planar*, with shape $(B, 3, L, N, N)$. The proposal takes the second. Under it, a run along x at fixed (b, a, z, y) — replica, component, layer, row — is contiguous in memory with unit stride, so a row of N spin components of one Cartesian component is a single contiguous segment. The transposition between layouts is performed once, outside the sweep loop, and therefore does not enter the per-sweep cost.

This component is a precondition for the next one: a shifted read is contiguous only if the unshifted read was.

2.3 A synchronisation-free halo: the shifted-load stencil

Let a thread block own a tile of B_y rows by B_x lanes of one layer. The stencil requires, for each of the three components, the tile itself and its four in-plane translates together with the two out-of-plane ones. The proposal obtains each translate as a *vector-shifted contiguous read*: the index expression is $x + 1$ applied to a contiguous run, so the access remains a contiguous segment displaced by one element and its contiguity is statically evident. The arithmetic of (5) is then carried out in registers and only the accepted configuration is written.

The halo requires *no barrier*. By the admissibility condition (6), every neighbour a block reads belongs to a colour class other than the one being updated, and is therefore written by no block during that pass. A block may read its halo from global memory with no coordination whatever. The colouring, introduced in Section 1.2 as a correctness device, is thus also what makes the locality safe — and the guarantee holds for any n_c, so it extends to the eight- and twenty-seven-class colourings unchanged.

2.4 Counter-based proposals: in-kernel generation

The trial spin and the acceptance variate are generated *inside* the kernel from a counter-based generator rather than supplied by the host. For site i of replica b at step n, let $\omega(b, i) = ((bL + z_i)N + y_i)N + x_i$ be the global site offset. The proposal draws

$$g_k = \mathcal{N}(n,\ 4\omega + k),\ k \in \{0, 1, 2\}, \qquad u = \mathcal{U}(n,\ 4\omega + 3), \qquad \mathbf{t}_i = S_0\, \frac{(g_0, g_1, g_2)}{\|(g_0, g_1, g_2)\|}, \tag{12}$$

with $\mathcal{N}$ and $\mathcal{U}$ the normal and uniform streams of the generator. The four draws share one key, the step index n, and are separated by *distinct counters* — the standard use of a counter-based generator. The separation is required rather than incidental: the acceptance variate u and the direction $\mathbf{t}_i$ must be statistically independent for the acceptance test of Section 1.2 to sample the intended chain, and distinct counters is what guarantees they share no input.

2.5 Deterministic observables: block-partial reduction

The observables of Section 2.6 below are accumulated as *per-block partial sums*, written to a small array and reduced afterwards. The reduction is therefore independent of the order in which blocks retire, so two evaluations of the same configuration agree to the last bit and the observables admit exact comparison. The partial array is of size $O(BL\lceil N/B_y\rceil)$ and its final reduction is negligible beside the sweep.

The topological charge is the one observable whose kernel is partial by construction: the Berg–Lüscher plaquette form requires a two-argument arctangent whose branch matters, since its denominator may change sign on a frustrated plaquette. The kernel emits the four arguments of the two arctangents and the final transcendental is applied outside it.

2.6 The observables, one by one

Six quantities are reported per temperature, and each is a reduction of the arrays of Section 2.1 over a stated set of axes with a stated normalisation. Let $n_a = |\Lambda|$ be the number of sites inside the disc and let χ_Λ be its indicator, the mask of (7). Every reduction below is masked by χ_Λ, so the sites that pad the box contribute nothing.

Energy per site, E. The total energy is evaluated with each bond counted *once*, using only the $+\hat{\mathbf{x}}$, $+\hat{\mathbf{y}}$ and $+\hat{\mathbf{z}}$ translates — never the six of V, which would count every bond twice:

$$E = \frac{1}{n_a} \sum_{i \in \Lambda} \Big[-\sum_{k \in \mathcal{S}} J_k \sum_{\mathbf{d} \in \mathcal{D}_k^+} \mathbf{s}_i \cdot \mathbf{s}_{i+\mathbf{d}} + \mathcal{A}_i(\mathbf{s}_i) - D\, \varepsilon_i^{\mathrm{DM}} - \mathbf{H} \cdot \mathbf{s}_i \Big], \tag{13}$$

where $\mathcal{D}_k^+$ is the half of $\mathcal{D}_k$ with positive displacement and $\varepsilon_i^{\mathrm{DM}}$ the in-plane cross-product term. The reduction is over the three spatial axes, leaving one value per replica, divided by n_a.

Magnetisation, $\|\mathbf{M}\|$ and M_z. The vector mean over in-disc sites,

$$\mathbf{M} = \frac{1}{n_a}\sum_{i\in\Lambda}\mathbf{s}_i, \qquad \text{reported as} \qquad \|\mathbf{M}\| = \Big(\textstyle\sum_a M_a^2\Big)^{1/2} \quad \text{and} \quad M_z = M_2. \tag{14}$$

Both are kept because they answer different questions: $\|\mathbf{M}\|$ cannot distinguish an in-plane from an out-of-plane texture, and M_z is the component the images of the later sections actually record.

Topological charge, Q. Computed on the mid-layer with the Berg–Lüscher plaquette form, which is quantised where a finite-difference curl is not. Each plaquette is split into two triangles and contributes the signed solid angle of each,

$$Q = \frac{1}{4\pi}\sum_{p\in\mathcal{P}}\Big[\Omega(\mathbf{s}_{00},\mathbf{s}_{10},\mathbf{s}_{11}) + \Omega(\mathbf{s}_{00},\mathbf{s}_{11},\mathbf{s}_{01})\Big], \qquad \Omega(\mathbf{a},\mathbf{b},\mathbf{c}) = 2\arctan\frac{\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})}{1+\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{c}+\mathbf{c}\cdot\mathbf{a}}, \tag{15}$$

where $\mathcal{P}$ is the set of plaquettes whose *four* corners lie in the disc — a plaquette straddling the boundary is dropped entirely rather than partially counted. Two details decide whether this is right. The arctangent must be the two-argument form, because the denominator changes sign on a frustrated plaquette and a one-argument arctan would silently pick the wrong branch; and the kernel therefore emits the four arguments of the two arctangents and leaves the transcendental to be applied outside it (Section 2.5).

Response functions, C_v and χ. Both are variances, and both are accumulated about a reference value E_{ref}, M_{ref} taken once at the start of the measurement phase of each temperature:

$$C_v = \frac{\langle(\Delta E)^2\rangle - \langle\Delta E\rangle^2}{k_B T^2 n_a}, \qquad \chi = \frac{\langle(\Delta\|\mathbf{M}\|)^2\rangle - \langle\Delta\|\mathbf{M}\|\rangle^2}{k_B T}, \tag{16}$$

with $\Delta E = E - E_{\mathrm{ref}}$ and likewise for M. A constant shift leaves a variance unchanged, so centring is exact; what it buys is that both terms of the subtraction sit at the scale of the fluctuation instead of the total. That is not a refinement but a requirement here: in single precision at low temperature, one unit in the last place of $\langle E^2\rangle$ exceeds the variance being measured, and the uncentred form returns a negative C_v.

What the kernel accumulates. The measurement phase therefore carries *eight* accumulators per replica and temperature, not six: the four centred moments $\sum\Delta E$, $\sum\Delta E^2$, $\sum\Delta\|\mathbf{M}\|$, $\sum\Delta\|\mathbf{M}\|^2$, then $\sum M_z$ and $\sum Q$, and finally the two references E_{ref} and M_{ref} needed to restore the absolute values. The reduction returns an array of shape $(n_T, 8, B)$ — temperatures first, the observable index second, replicas last — and the replica axis is averaged last, after the six reported quantities have been formed.

2.7 The combination

One sweep applies the kernel of Sections 2.2–2.4 once per colour class, n_c times; one temperature applies the sweep n_{th} times to thermalise and n_{me} times while accumulating the observables of Section 2.6; and one simulation scans a decreasing temperature ladder $\{T_m\}_{m=1}^{M}$, carrying the final configuration of each temperature into the next:

$$\mathbf{s}^{(m)} = \Big(\bigcirc_{c=n_c-1}^{0}\mathcal{K}_{\beta_m}^{(c)}\Big)^{n_{\mathrm{th}}+n_{\mathrm{me}}}(\mathbf{s}^{(m-1)}), \qquad \beta_m = 1/k_B T_m, \tag{17}$$

where $\bigcirc$ denotes composition over the colour classes in descending order, $\mathcal{K}_\beta^{(c)}$ is the single-pass kernel for class c at inverse temperature β, and the whole composition is evaluated for the B replicas at once. Every symbol traces to a declaration: $\mathcal{K}^{(c)}$ to Section 2.3, the classes c and their count n_c to (6), β to Section 1.2, and B to Section 2.2.

Figure 1 summarises the four components in the order this section defines them, showing for one pass how the layout, the halo, the generator and the reduction compose into the kernel.

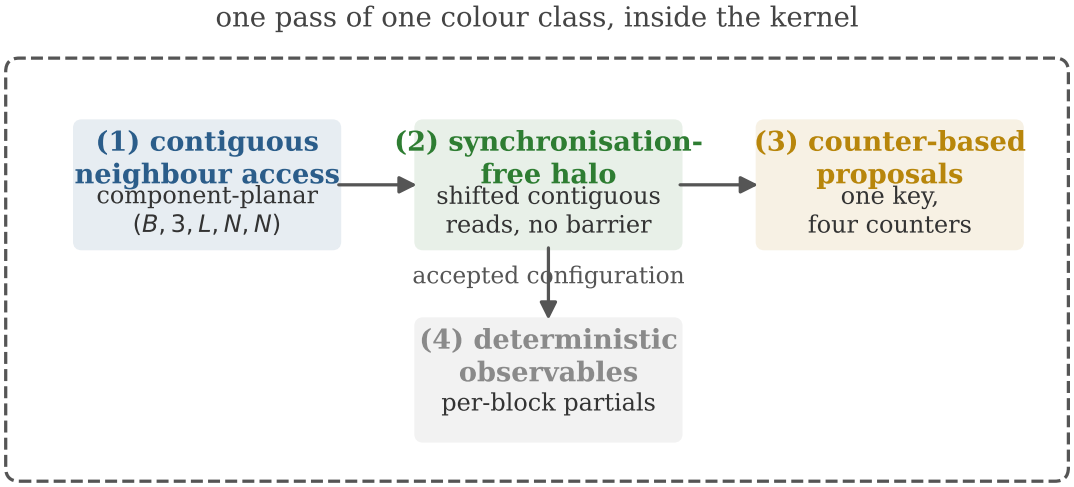


Figure 1: The fused single-pass kernel. (1) *Contiguous neighbour access*: the configuration is held component-planar, so a row of one Cartesian component is a contiguous segment. (2) *A synchronisation-free halo*: each block reads its tile and the one-site border through shifted contiguous reads, with no barrier, because the halo is of the opposite colour. (3) *Counter-based proposals*: the trial direction and the acceptance variate are drawn in-kernel from one key and four distinct counters. (4) *Deterministic observables*: per-block partial sums are written to a small array and reduced outside the kernel. Colour marks the component a box belongs to; the dashed boundary is the kernel.

3 Validation

3.1 The simulator samples the intended chain

Table 2 collects the four exactness tests and the three statistical ones, with the number each returns. Fed identical randomness, the fused kernel of Section 2.3 and the reference implementation of the chain agree on every spin of the configuration (0 of 182 520 differing, Table 2 row 1), with the single-site energy difference matching to 1.4×10^{-6} in absolute value, which is the resolution of single precision on that quantity. The observable kernels of Section 2.5 agree with the reference observables on a common configuration to 1.2×10^{-7} relative for the energy and 8.4×10^{-8} absolute for the topological charge (rows 3 and 4). These rows carry no statistics: with the randomness held fixed the only admissible difference is arithmetic, so a non-zero spin count would identify a term of (2) wrongly transcribed rather than a sampling discrepancy.

Row 7 is the instrument that settles the question, and reading it requires the control the row names. Running the reference implementation against *itself* with a second seed returns 0.87 and 2.53 for the same two statistics, which is the noise floor of the comparison rather than a property of either engine; the fused kernel's 0.99 and 1.90 therefore sit inside that floor, its maximum deviation falling below the control's own. Without that control a figure in this row is not interpretable in either direction, because the two engines draw from different generators and are therefore distinct Markov chains that can agree only in distribution.

Table 2: Agreement with the reference implementation of the chain. Rows 1–4 hold the randomness fixed and are exactness tests; rows 5–7 let each engine draw its own and are statistical.

#	test	quantity	result
1	one pass, even class, identical randomness	differing spins	0/182 520
2	one pass, odd class, identical randomness	differing spins	0/182 520
3	observable kernels, energy	relative	1.2×10^{-7}
4	observable kernels, topological charge	absolute	8.4×10^{-8}
5	both chains to equilibrium, $T = 20, 5, 1$	σ	$0.09 - 0.23$
6	trial direction norm	deviation from S_0	2.4×10^{-7}
7	reduced curves, reference run against itself	mean σ / max σ	0.99/1.90

The admissibility condition (6) is checked computationally for the colouring in use rather than argued from Table 1. Over the displacement sets of the active shells, the two-class colouring is admissible for shells $\{1\}$ and $\{1, 3\}$, the eight-class colouring for any set containing shell 2, and the twenty-seven-class colouring for any set containing shell 4; in every case the colour classes partition the sites inside the disc exactly. Because the no-barrier guarantee of Section 2.3 follows from that condition and not from the class count, it holds unchanged at eight and twenty-seven classes.

3.2 Each component delivers the property it was defined under

3.2.1 Contiguous neighbour access and the synchronisation-free halo

Table 3: Time per single-site update, per device, all taken at five thousand replicas per device so the rows share a basis. The processor baseline runs the same code, the same Hamiltonian and the same workload, which is what makes it a controlled comparison rather than a quotation.

configuration	device	ns / update
this work, one device	H100	**0.129**
this work, one device	T4	1.75
processor baseline	24 threads	107.7

Table 3 reports the per-site update time the two components jointly achieve: 0.129 ns on one datacentre GPU (Table 3, row 1), and 0.033 ns aggregated over four, which that table deliberately omits because it tabulates per-device rates only. At the geometry of Section 1.1 that is 7.74 site updates per nanosecond per device. The mechanism is the one the layout was defined for: the weighted neighbour sum (5) is formed from contiguous segments displaced by one element, so each translate costs one coalesced transaction per segment rather than a gather, and no translate is ever written to memory.

3.2.2 Counter-based proposals

Table 4 presents the acceptance rate of the in-kernel generator against the reference generator at five temperatures spanning the ladder. The ratio scatters on both sides of unity (0.996 to 1.029, Table 4 column 4), each value within one standard error of it, where the standard error of a ratio of two independently measured acceptance rates is $\sqrt{2}\,\sqrt{(1-p)/(Np)}$ and runs from 0.47 % at the hottest temperature to 3.4 % at the coldest, the acceptance rate p falling by two orders of magnitude across that range. The absence of a systematic sign **confirms** the independence that Section 2.4 requires between the acceptance variate and the trial direction: a shared input between them would bias the test in one direction at every temperature, which is the signature this instrument is built to detect.

Table 4: Acceptance rate of the in-kernel generator against the reference generator, from a common configuration. The expected ratio is unity; the standard error grows as the acceptance rate falls.

T (K)	reference	in-kernel	ratio	std. error (%)
20.0	0.4233	0.4218	0.996	0.47
10.0	0.1614	0.1611	0.998	0.97
5.0	0.0723	0.0718	0.994	1.50
1.0	0.0132	0.0136	1.029	3.40
0.1	0.0029	0.0029	0.997	7.30

3.2.3 Deterministic observables

Rows 3 and 4 of Table 2 are also this component's evidence, and they are available *as* exactness tests only because the reduction is order-independent: a reduction whose result depended on the order in which blocks retired could be compared only within a tolerance, and the figures of 1.2×10^{-7} and 8.4×10^{-8} would carry no information about the formulation. That the comparison is admissible at all is **attributed to** the partial-sum reduction of Section 2.5 rather than inferred from the agreement.

4 Performance

4.1 At production scale

Figure 2 is the principal result. It presents the wall clock of the complete production ladder — 200 temperatures by 20 000 sweeps, so 4×10^6 sweeps per replica — for one hundred states at five thousand replicas, across the CPU baseline, one and two consumer GPUs and one, two and four datacentre GPUs.

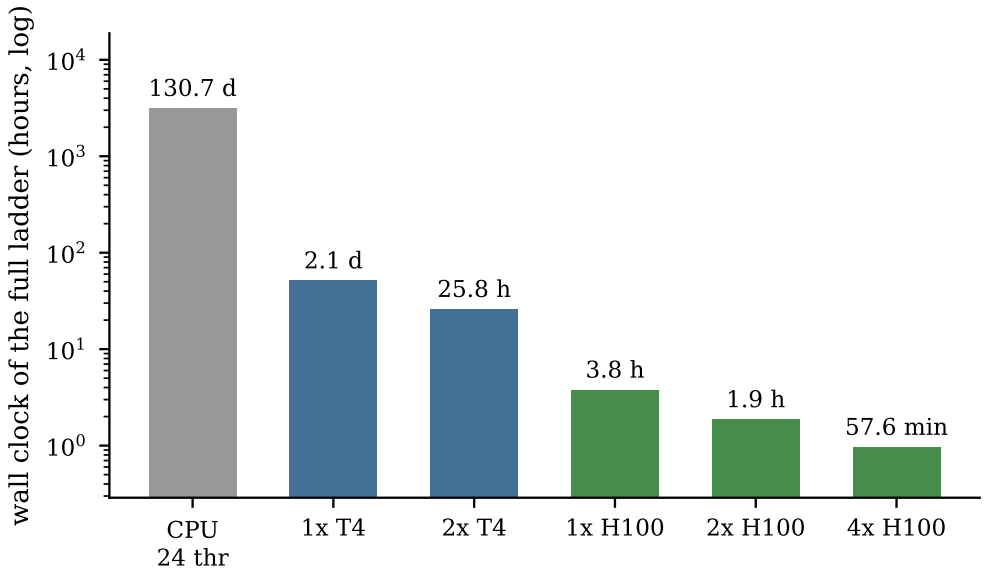


Figure 2: The production ladder for one hundred states at five thousand replicas: 200 temperatures by 20 000 sweeps, 4×10^6 sweeps per replica. Wall clock on a logarithmic axis, since at 130.7 d the processor baseline compresses every other bar onto the axis.

Read at four datacentre GPUs (Figure 2, rightmost bar), the simulator completes the hundred-state ladder in 57.6 min; at one device (Figure 2, fourth bar) in 3.8 h; and on two consumer GPUs (Figure 2, third bar) in 25.8 h. Against the CPU baseline on the identical

workload, 130.7 d, the separation is 834-fold per site update (Table 3, rows 1 and 3). Figure 3 shows the same comparison without the CPU bar and on a linear axis.

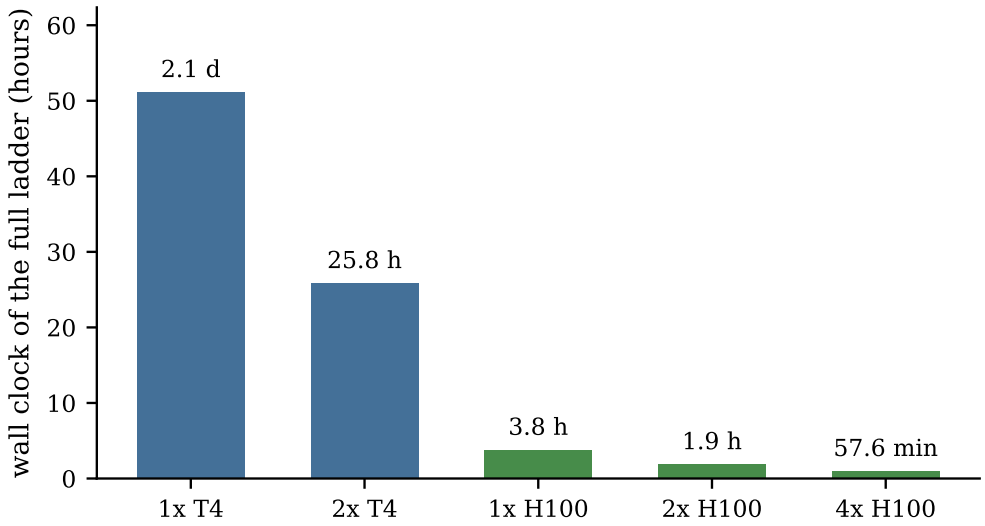


Figure 3: The same ladder as Figure 2 with the processor baseline removed and a linear axis, which is what makes the separation among the accelerated configurations legible.

The hundred-state configuration is the one that exhibits the scaling, which is why it is the principal figure rather than the single-state case. Over one to four devices the simulator gains a factor of 3.91 with a hundred states and only 1.10 with one, on the same hardware and the same kernel. This is **attributed to** the replicas available per device rather than inferred from the two figures: at one state and four devices each device receives twelve replicas, and Figure 4 shows the per-site cost against replicas per device, where the curve rises steeply below roughly twenty-five. The practical consequence is that additional devices are paid for by additional states — one state per device requires no sharding and preserves the per-state cost — and not by dividing one state across them.

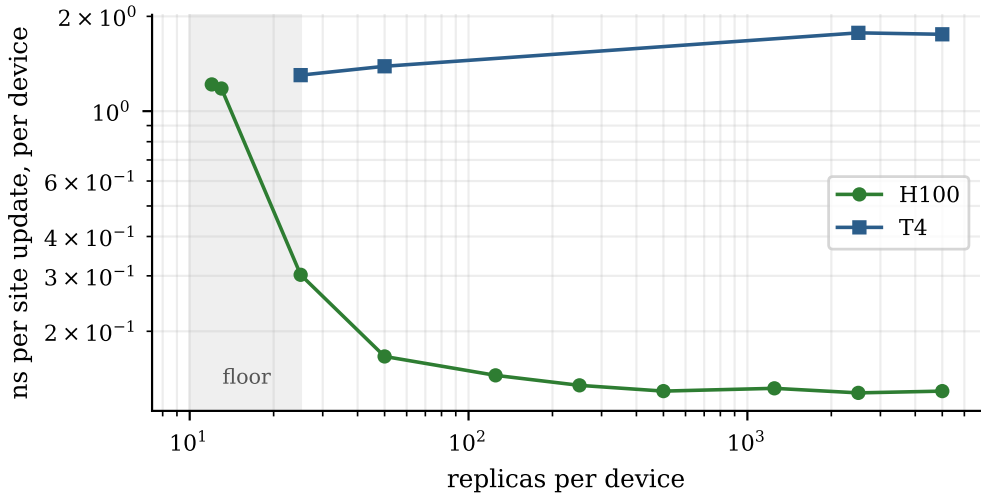


Figure 4: Per-site update cost against replicas per device, at fixed geometry. Below roughly twenty-five replicas per device the curve rises steeply: this is the floor that makes a single state fail to use four devices. The datacentre device improves as the batch widens and then plateaus; the consumer device degrades monotonically over the same range.

Figure 5 presents the per-site cost against the replica count at fixed geometry and shows it very nearly insensitive to the arithmetic form of the array extent. Let v_2 be the 2-adic

valuation, B the replicas per device and L the layer count. The condition $v_2(B) + v_2(L) \geq 2$ can be flipped by varying B only when $v_2(L) < 2$, which among the layer counts measured holds for $L = 5$ and $L = 6$ and not for $L = 4$ or $L = 8$, where the condition is already satisfied for every B. Across the two informative layer counts the condition changes the cost by 0.30 % and 1.70 % on the datacentre device, and by 0.56 % and 0.97 % on the consumer device — at most 1.7 %, with no consistent sign. This is **attributed to** the formulation: the block shape is fixed by the kernel of Section 2.3 and the layer count only sets how many blocks exist along the out-of-plane axis, so no code-generation decision depends on the extent. The practical consequence is that the replica count may be chosen for the physics or for the scheduling.

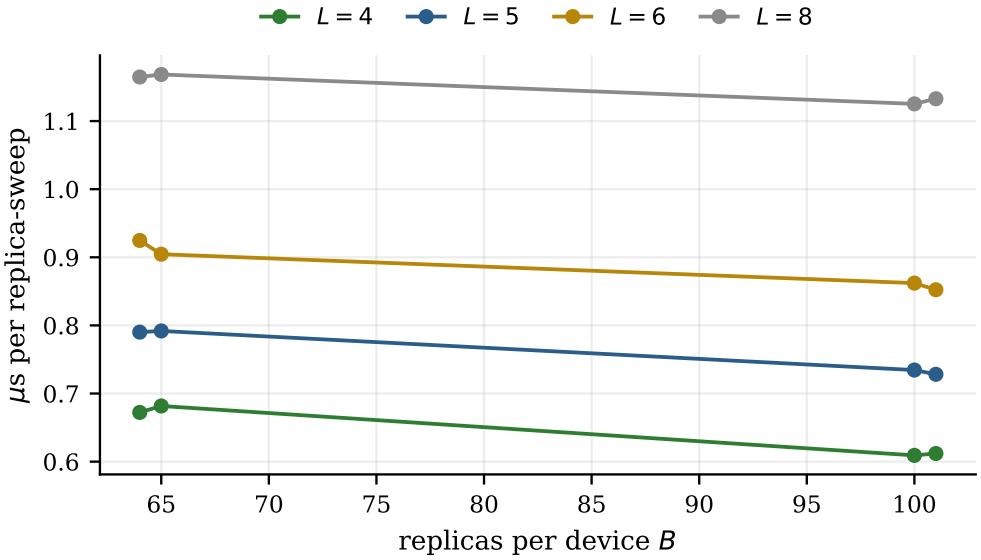


Figure 5: Per-replica-sweep cost against the replica count at fixed geometry, one line per layer count, on the datacentre device. Each pair $B = 64, 65$ and $B = 100, 101$ flips the condition $v_2(B) + v_2(L) \geq 2$ at $L = 5$ and $L = 6$; the lines cross those pairs flat, by 0.30 % and 1.70 %. At $L = 4$ and $L = 8$ the condition holds for every B, so those two lines carry no information about it and are shown for the cost scale.

The margin is not uniform across devices, and Figure 4 shows the sign. On the datacentre device the per-site cost improves as the batch widens and then plateaus at 0.677 µs per replica-sweep; on the consumer device it degrades monotonically, from 7.282 to 9.201 µs over the same range. This **suggests** a memory-hierarchy origin — the consumer device has an order of magnitude less last-level cache and a third of the bandwidth, against a working set of 456 MB at five thousand replicas — but that mechanism was not measured here, and only the sign is established.

5 Limitations

Four limitations bound the claims above. The per-device replica floor, below which the per-site cost rises steeply (Figure 4). The architecture dependence of the margin, whose sign is measured and whose mechanism is not. The widening of the colouring that shells 2 and 4 require, which divides the parallelism available per pass by eight or twenty-seven while leaving the work per sweep unchanged, and which the fused kernel of Section 2.3 does not yet implement beyond the first shell. And the single geometry in which the cost behaviour of Figure 5 was characterised.